

## CHALLENGER EXERCISES

### Exercise C - I

#### SINGLE CHOICE TYPE QUESTIONS

1. If one root of the quadratic equation  $ax^2 + bx + c = 0$  be  $n^{\text{th}}$  root of the other then  $(ac^n)^{1/n+1} + (a^n c)^{1/n+1} =$   
यदि द्विघात समीकरण  $ax^2 + bx + c = 0$  का एक मूल दूसरे कि  $n^{\text{th}}$  घात हो,  $(ac^n)^{1/n+1} + (a^n c)^{1/n+1} =$   
(A)  $b^{1/n+1}$  (B)  $b^{1/n}$  (C)  $b$  (D)  $-b$

Ans. D

Sol.  $\alpha \cdot \alpha^n = \frac{c}{a} \Rightarrow \alpha = \left(\frac{c}{a}\right)^{\frac{1}{n+1}} \Rightarrow \alpha + \alpha^n = -\frac{b}{a} \Rightarrow \left(\frac{c}{a}\right)^{\frac{1}{n+1}} + \left(\frac{c}{a}\right)^{\frac{n}{n+1}} = -\frac{b}{a}$   
 $a^{\frac{1}{n+1}} \cdot c^{\frac{1}{n+1}} + c^{\frac{n}{n+1}} \cdot a^{\frac{1}{n+1}} + b = 0 \Rightarrow a^{\frac{n}{n+1}} \cdot c^{\frac{1}{n+1}} + a^{\frac{1}{n+1}} \cdot c^{\frac{n}{n+1}} + b = 0$   
 $(a^n c)^{\frac{1}{n+1}} + (a c^n)^{\frac{1}{n+1}} + b = 0$  Proved.

2. If  $\alpha, \beta$  are the roots of the equation  $ax^2 + bx + c = 0$  and  $S_n = \alpha^n + \beta^n$ , then  $a S_{n+1} + c S_{n-1} =$   
यदि  $\alpha, \beta$  समीकरण  $ax^2 + bx + c = 0$  के मूल हो तथा  $S_n = \alpha^n + \beta^n$  तथा  $a S_{n+1} + c S_{n-1} =$   
(a)  $b S_n$  (B)  $b^2 S_n$  (C)  $2b S_n$  (D)  $-b S_n$

Ans. D

Sol.  $ax^2 + bx + c = 0 \begin{cases} \alpha \\ \beta \end{cases}; S_n = \alpha^n + \beta^n$   
 $a\alpha^2 + b\alpha + c = 0$   
 $\Rightarrow a\alpha^2 + c = -b\alpha$   
 $a\beta^2 + b\beta + c = 0$   
 $a\beta^2 + c = -b\beta$   
Now  $aS_{n+1} + cS_{n-1}$   
 $= a(\alpha^{n+1} + \beta^{n+1}) + c(\alpha^{n-1} + \beta^{n-1})$   
 $= a\alpha^{n+1} + c\alpha^{n-1} + a\beta^{n+1} + c\beta^{n-1}$   
 $= \alpha^n \left( a\alpha + \frac{c}{\alpha} \right) + \beta^n \left( a\beta + \frac{c}{\beta} \right)$   
 $= \frac{\alpha^n}{\alpha} (a\alpha^2 + c) + \frac{\beta^n}{\beta} (a\beta^2 + c)$

$$= \frac{\alpha^n}{\alpha}(b\alpha) + \frac{\beta^n}{\beta}(-b\beta)$$

$$= -b(\alpha^n + \beta^n) = -bS_n$$

3. If the equation  $x^2 + 2(k+1)x + 9k - 5 = 0$  has only negative roots, then :

यदि समीकरण  $x^2 + 2(k+1)x + 9k - 5 = 0$  के केवल ऋणात्मक मूल हों, तो :

(A)  $k \leq 0$

(B)  $k \geq 0$

(C)  $k \geq 6$

(D)  $k \leq 6$

**Ans.** C

**Sol.** Equation  $x^2 + 2(k+1)x + (9k-5) = 0$

$\therefore$  Roots only negative

(i)  $\therefore D \geq 0$

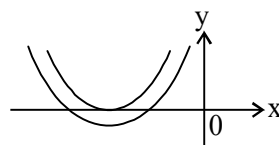
$$4(k+1)^2 - 4(9k-5) \geq 0$$

$$k^2 + 1 + 2k - 9k + 5 \geq 0$$

$$k^2 - 7k + 6 \geq 0$$

$$(k-6)(k-1) \geq 0$$

$$k \leq 1, k \geq 6$$



(ii)  $-\frac{b}{2a} < 0$

$$k \in (-1, \infty)$$

(iii)  $f(0) > 0$

$$k \in \left(\frac{5}{9}, \infty\right)$$

(i)  $\cap$  (ii)  $\cap$  (iii)

$$k \in [6, \infty)$$

4. The set of values of 'a' for which  $f(x) = ax^2 + 2x(1-a) - 4$  is negative for exactly three integral values of x, is :

'a' के मानों का समुच्चय जिसके लिये  $f(x) = ax^2 + 2x(1-a) - 4$  के ठीक तीन पूर्णांक मानों के लिये ऋणात्मक है, होगा :

(A)  $(0, 2)$

(B)  $(0, 1]$

(C)  $[1, 2)$

(D)  $[2, \infty)$

**Ans.** C

**Sol.**  $f(x) = ax^2 + 2x(1-a) - 4$

$$x = \frac{-2(1-a) \pm \sqrt{4(1-a)^2 + 16a}}{2a}$$

$$x = \frac{+2(a-1) \pm 2\sqrt{1+a^2-2a+4a}}{2a}$$

$$x = \frac{(a-1) \pm \sqrt{(1+a)^2}}{a}$$

$$x = \frac{a-1+1+a}{a} = \frac{2a}{a} = 2,$$

$$\text{and } x = \frac{a-1-1-a}{a}$$

$$x = -\frac{2}{a}$$

$$-\frac{2}{a} < -1, \frac{-2}{a} \geq -2$$

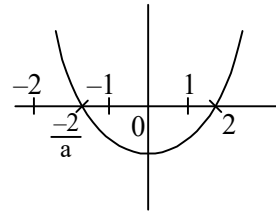
$$\frac{-2}{a} + 1 < 0, -\frac{2}{a} + 2 \geq 0$$

$$\frac{a-2}{a} < 0, \frac{2a-2}{a} \geq 0$$

$$a < 2, \frac{2(a-1)}{a} \geq 0$$

$$a \geq 1$$

$$\therefore a \in [1, 2)$$



5. The number of real solutions of the equation  $(15 + 4\sqrt{14})^t + (15 - 4\sqrt{14})^t = 30$  are, where  $t = x^2 - 2|x|$  :

समीकरण  $(15 + 4\sqrt{14})^t + (15 - 4\sqrt{14})^t = 30$  के वास्तविक हलों की संख्या क्या होगी, जहाँ  $t = x^2 - 2|x|$  :

- (A) 0 (B) 2 (C) 4 (D) 6

**Ans.** C

**Sol.**  $(15 + 4\sqrt{14})^t + (15 - 4\sqrt{14})^t = 30$

Let  $u = 15 + 4\sqrt{14}$

$$\frac{1}{u} = 15 - 4\sqrt{14}$$

$$u^t + \frac{1}{u^t} = 30$$

put  $u^t = p$

$$p + \frac{1}{p} = 30$$

$$p^2 - 30p + 1 = 0$$

$$p = \frac{30 \pm \sqrt{900 - 4}}{2}$$

$$p = \frac{30 \pm \sqrt{896}}{2}$$

$$p = \frac{30 \pm 8\sqrt{14}}{2}$$

$$p = 15 \pm 4\sqrt{14}$$

$$p = 15 + 4\sqrt{14}, 15 - 4\sqrt{14}$$

$$u^t = 15 + 4\sqrt{14}, u^t = 15 - 4\sqrt{14}$$

$$(15 + 4\sqrt{14})^t = (15 + 4\sqrt{14})^1$$

$$t = 1$$

And

$$(15 + 4\sqrt{14})^t = 15 - 4\sqrt{14}$$

$$(15 + 4\sqrt{14})^{t+1} = (15 - 4\sqrt{14}) \cdot (15 + 4\sqrt{14})$$

$$(15 + 4\sqrt{14})^{t+1} = 1$$

$$t + 1 = 0$$

$$t = -1$$

for  $t = 1$

$$x^2 - 2|x| = 1$$

$$\text{for } x \geq 0 \quad ;$$

$$x^2 - 2x - 1 = 0 \quad ;$$

$$x = \frac{+1 \pm \sqrt{2}}{1} \quad ;$$

$$x = +1 + \sqrt{2}, +1 - \sqrt{2}$$

for  $t = -1$

$$x^2 - 2|x| = -1$$

$$x^2 - 2|x| + 1 = 0$$

$$\text{for } |x| < 0 \quad ;$$

$$x^2 + 2x + 1 = 0 \quad ;$$

$$(x + 1)^2 = 0 \quad ;$$

$$x = -1 \quad ;$$

Total no. of solutions = 4

for  $x < 0$

$$x^2 + 2x - 1 = 0$$

$$x = -1 + \sqrt{2}, -1 - \sqrt{2}$$

for  $x \geq 0$

$$x^2 - 2x + 1 = 0$$

$$(x - 1)^2 = 0$$

$$x = 1$$

6. If  $\alpha, \beta$  are the roots of  $ax^2 + bx + c = 0$ , ( $a \neq 0$ ), and  $\alpha + \delta, \beta + \delta$  are the roots of  $Ax^2 + Bx + C = 0$ , ( $A \neq 0$ ) for some constant  $\delta$ , then the value of  $\frac{b^2 - 4ac}{a^2}$  is :

(A)  $\frac{B^2 - 4AC}{A^2}$  (B)  $\frac{B^2 + 4AC}{A^2}$  (C)  $\frac{A^2 - 4BC}{A^2}$  (D) None of these

यदि  $\alpha, \beta$  समीकरण  $ax^2 + bx + c = 0$ , ( $a \neq 0$ ) के मूल व किसी अचर राशि  $\delta$  के लिए  $\alpha + \delta, \beta + \delta$  समीकरण

$Ax^2 + Bx + C = 0$ , ( $A \neq 0$ ) के मूल हों, तो  $\frac{b^2 - 4ac}{a^2}$  का मान होगा :

(A)  $\frac{B^2 - 4AC}{A^2}$  (B)  $\frac{B^2 + 4AC}{A^2}$  (C)  $\frac{A^2 - 4BC}{A^2}$  (D) इनमें से कोई नहीं

**Ans.** A

**Sol.**  $ax^2 + bx + c = 0 \begin{matrix} \alpha \\ \beta \end{matrix} \Rightarrow \alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a} \Rightarrow Ax^2 + Bx + C = 0 \begin{matrix} \alpha + \delta \\ \beta + \delta \end{matrix}$

$(\alpha + \delta) + (\beta + \delta) = -\frac{B}{A}, (\alpha + \delta)(\beta + \delta) = \frac{C}{A} \therefore |(\alpha + \delta) - (\beta + \delta)| = |(\alpha - \beta)|$

$\Rightarrow \sqrt{\frac{B^2}{A^2} - \frac{4C}{A}} = \sqrt{\frac{b^2}{a^2} - \frac{4c}{a}} \Rightarrow \frac{b^2 - 4ac}{a^2} = \frac{B^2 - 4AC}{A^2}$  Hence proved

7. Given  $a, b, c$  are non negative real numbers and if  $a^2 + b^2 + c^2 = 1$ , then the value of  $a + b + c$  is :

दिया गया है कि  $a, b, c$  अऋणात्मक वास्तविक संख्यायें हैं तथा यदि  $a^2 + b^2 + c^2 = 1$  है तो  $a + b + c$  का मान होगा :

(A)  $\geq 3$  (B)  $\geq 2$  (C)  $\leq \sqrt{2}$  (D)  $\leq \sqrt{3}$

**Ans.** D

**Sol.**  $(a + b + c)^2 = a^2 + b^2 + c^2 + 2 \sum ab = 1 + 2 \sum ab$

put  $a^2 + b^2 + c^2 \geq ab + bc + ca$

or  $\sum ab \leq 1$

$\therefore (a + b + c)^2 \leq 1 + 2$

$\therefore (a + b + c) \leq \sqrt{3}$

Alternatively : using  $RMS \geq AM$  in  $a, b, c$

$\sqrt{\frac{a^2 + b^2 + c^2}{3}} \geq \frac{a + b + c}{3} \Rightarrow \sqrt{3} \geq (a + b + c)^2 \Rightarrow a + b + c \leq \sqrt{3}$

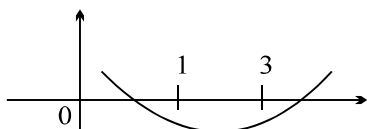
8. The set of values of 'a' for which the inequality,  $(x - 3a)(x - a - 3) < 0$  is satisfied for all  $x \in [1, 3]$  is :

'a' के मानों का समुच्चय जिनके लिये असमिका  $(x - 3a)(x - a - 3) < 0$  सभी  $x \in [1, 3]$  के लिये संतुष्ट होती है, क्या होगा :

(A)  $(1/3, 3)$  (B)  $(0, 1/3)$  (C)  $(-2, 0)$  (D)  $(-2, 3)$

**Ans.** B

**Sol.** Equation is  $x^2 - (4a + 3)x + 3a(a + 3)$



$$f(1) < 0 \text{ and } f(3) < 0$$

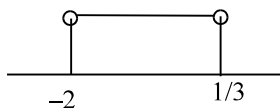
$$(1 - 3a)(1 - a - 3) < 0 \Rightarrow 1 - a - 3 - 3a + 3a^2 + 9a < 0$$

$$\Rightarrow 3a^2 + 5a - 2 < 0$$

$$3a^2 + 6a - a - 2 < 0$$

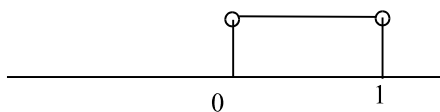
$$3a(a + 2) - (a + 2) < 0$$

$$(a + 2)(3a - 1) < 0 \Rightarrow$$



$$\text{again } (3 - 3a)(-a) < 0 \Rightarrow (a - 1)a < 0$$

$$\Rightarrow 0 < a < 1$$



$$\text{Hence } a \in \left(0, \frac{1}{3}\right)$$

9. If  $\alpha$  and  $\beta$  are the roots of  $x^2 + px + q = 0$  and  $\alpha^4, \beta^4$  are the roots of  $x^2 - rx + s = 0$ , then the equation  $x^2 - 4qx + 2q^2 - r = 0$  has always  $(p, q, r, s \in \mathbb{R})$  :

(A) two real roots

(B) two positive roots

(C) two negative roots

(D) one positive and one negative root

यदि समीकरण  $x^2 + px + q = 0$  के मूल  $\alpha$  तथा  $\beta$  हैं तथा समीकरण  $x^2 - rx + s = 0$  के मूल  $\alpha^4, \beta^4$  हैं तो समीकरण  $x^2 - 4qx + 2q^2 - r = 0$  के सदैव  $(p, q, r, s \in \mathbb{R})$  :

(A) दो वास्तविक मूल होंगे

(B) दो धनात्मक मूल होंगे

(C) दो ऋणात्मक मूल होंगे

(D) एक धनात्मक तथा एक ऋणात्मक मूल होगा

**Ans.** A

$$\text{Sol. } x^2 + px + q = 0 \begin{cases} \alpha \\ \beta \end{cases} ;$$

$$\alpha^4 + \beta^4 = (\alpha^2 + \beta^2)^2 - 2\alpha^2\beta^2$$

$$\alpha + \beta = -p, \alpha\beta = q ;$$

$$r = [(\alpha + \beta)^2 - 2\alpha\beta]^2 - 2\alpha^2\beta^2$$

$$\text{And } x^2 - rx + q = 0 ;$$

$$r = [p^2 - 2q]^2 - 2q^2$$

$$\alpha^4 + \beta^4 = r = +ve$$

Now the equation :  $x^2 - 4qx + 2q^2 - r = 0$

$$D = 16q^2 - 4(2q^2 - r)$$

$$= 16q^2 - 8q^2 + 4r$$

$$= 8q^2 + 4r = 8q^2 + 4 \left[ (p^2 - 2q)^2 - 2q^2 \right]$$

$$= 8q^2 + 4(p^2 - 2q)^2 - 8q^2$$

$$= 4(p^2 - 2q)^2$$

$$= +ve$$

$\therefore$  Equation have two real roots.

10. Number of integral values of 'a' for which the equation  $x^2 - (a+1)x + a - 1 = 0$ , has integral roots, is equal to :

'a' के पूर्णांक मानों की संख्या समीकरण  $x^2 - (a+1)x + a - 1 = 0$  के लिये पूर्णांक मूल रखती है, बराबर है :

(A) 1

(B) 2

(C) 3

(D) 4

**Ans.** A

**Sol.**  $x^2 - (a+1)x + (a-1) = 0$

$$D = \text{per. sq.}$$

$$D = P^2$$

$$\text{Let } P = 2$$

$$(a+1)^2 - 4(a-1) = 4$$

$$a^2 + 1 + 2a - 4a + 4 = 4$$

$$(a^2 + 1 - 2a) = 0$$

$$(a-1)^2 = 0$$

$$a = 1$$

There is only one case.

$\therefore$  Integral value of  $a = 1$ .

11. Let  $a, b, c$  be the sides of a triangle. No two of them are equal and  $\lambda \in \mathbb{R}$ . If the roots of the equation  $x^2 + 2(a+b+c)x + 3\lambda(ab+bc+ca) = 0$  are real, then :

माना  $a, b, c$  त्रिभुज की भुजाएँ हैं, जिनमें से कोई भी दो समान नहीं हैं एवं  $\lambda \in \mathbb{R}$  है। यदि समीकरण  $x^2 + 2(a+b+c)x + 3\lambda(ab+bc+ca) = 0$  के मूल वास्तविक हैं तो :

(A)  $\lambda < \frac{4}{3}$

(B)  $\lambda > \frac{5}{3}$

(C)  $\lambda \in \left(\frac{1}{3}, \frac{5}{3}\right)$

(D)  $\lambda \in \left(\frac{4}{3}, \frac{5}{3}\right)$

**Ans.** A

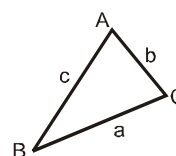
**Sol.** We, know that  $a+b > c, b+c > a$  and  $c+a > b \Rightarrow c-a < b, a-b < c, b-c < a$

squaring on both sides and adding  $(c-a)^2 + (a-b)^2 + (b-c)^2 < a^2 + b^2 + c^2$

$$a^2 + b^2 + c^2 - 2(ab+bc+ca) < 0 \Rightarrow (a+b+c)^2 - 4(ab+bc+ca) < 0$$

$$\Rightarrow \frac{(a+b+c)^2}{ab+bc+ca} < 4$$

\_\_\_\_\_ (i)



Now roots of equation  $x^2 + 2(a + b + c)x + 3\lambda(ab + bc + ca) = 0$  are real, then  $D \geq 0$

$$\Rightarrow 4(a + b + c)^2 - 4 \cdot 3\lambda(ab + bc + ca) \geq 0 \Rightarrow \frac{(a + b + c)^2}{ab + bc + ca} \geq 3\lambda$$

$$\text{So } 3\lambda \leq \frac{(a + b + c)^2}{ab + bc + ca} < 4 \Rightarrow \lambda < \frac{4}{3}$$

12. If  $\alpha, \beta$  are the roots of equation  $x^2 + x - 2 = 0$ , then the value of  $\frac{\alpha\beta^4(\beta+1)^4 + \beta\alpha^4(\alpha+1)^4}{\alpha^2 + \beta^2 + \alpha + \beta}$  is equal to :

यदि  $\alpha, \beta$  समीकरण  $x^2 + x - 2 = 0$  के मूल हैं, तो  $\frac{\alpha\beta^4(\beta+1)^4 + \beta\alpha^4(\alpha+1)^4}{\alpha^2 + \beta^2 + \alpha + \beta}$  का मान है :

- (A) 2 (B) -2 (C) 4 (D) -4

**Ans.** D

**Sol.**  $x^2 + x - 2 = 0$   $\alpha + \beta = -1, \alpha\beta = -2, \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 1 + 4 = 5$

$$\alpha^2 + \alpha - 2 = 0 \quad \beta^2 + \beta - 2 = 0$$

$$\alpha(\alpha + 1) = 2 \quad \beta(\beta + 1) = 2$$

$$\alpha + 1 = \frac{2}{\alpha} \quad \beta + 1 = \frac{2}{\beta}$$

Now,  $\frac{\alpha\beta^4(\beta+1)^4 + \beta\alpha^4(\alpha+1)^4}{\alpha^2 + \beta^2 + \alpha + \beta}$

$$= \frac{\alpha\beta^4 \cdot \frac{2^4}{\beta^4} + \beta\alpha^4 \cdot \frac{2^4}{\alpha^4}}{5 - 1}$$

$$= \frac{16(\alpha + \beta)}{4} = 4 \times (-1) = -4$$

13. If  $\alpha, \beta, \gamma$  are the roots of the cubic  $x^3 - 2x + 3 = 0$  then the value of  $\frac{1}{\alpha^3 + \beta^3 + 6} + \frac{1}{\beta^3 + \gamma^3 + 6} + \frac{1}{\gamma^3 + \alpha^3 + 6}$  equals :

यदि  $\alpha, \beta, \gamma$  घनीय समीकरण  $x^3 - 2x + 3 = 0$  के मूल हैं, तो  $\frac{1}{\alpha^3 + \beta^3 + 6} + \frac{1}{\beta^3 + \gamma^3 + 6} + \frac{1}{\gamma^3 + \alpha^3 + 6}$  का मान है :

- (A)  $\frac{1}{3}$  (B)  $-\frac{1}{3}$  (C)  $\frac{1}{2}$  (D)  $-\frac{1}{2}$

**Ans.** B

**Sol.**  $x^3 - 2x + 3 = 0 \begin{cases} \alpha \\ \beta \\ \gamma \end{cases}$

$$\alpha + \beta + \gamma = 0$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = -2$$

$$\alpha + \beta = -\gamma, \quad \beta + \gamma = -\alpha, \quad \gamma + \alpha = -\beta$$

$$\alpha\beta\gamma = -3$$

$$\Rightarrow \alpha^3 - 2\alpha + 3 = 0$$

$$\alpha^3 + 3 = 2\alpha$$

Similarly

$$\beta^3 + 3 = 2\beta, \quad \gamma^3 + 3 = 2\gamma$$

$$\begin{aligned} \text{Now} \quad & \frac{1}{\alpha^3 + \beta^3 + 6} + \frac{1}{\beta^3 + \gamma^3 + 6} + \frac{1}{\gamma^3 + \alpha^3 + 6} \\ &= \frac{1}{(\alpha^3 + 3) + (\beta^3 + 3)} + \frac{1}{(\beta^3 + 3) + (\gamma^3 + 3)} + \frac{1}{(\gamma^3 + 3) + (\alpha^3 + 3)} \\ &= \frac{1}{(2\alpha + 2\beta)} + \frac{1}{(2\beta + 2\gamma)} + \frac{1}{(2\gamma + 2\alpha)} \\ &= \frac{1}{2(\alpha + \beta)} + \frac{1}{2(\beta + \gamma)} + \frac{1}{2(\gamma + \alpha)} \\ &= \frac{1}{-2\gamma} + \frac{1}{-2\alpha} + \frac{1}{-2\beta} \\ &= -\frac{1}{2} \left[ \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} \right] \\ &= -\frac{1}{2} \times \frac{-2}{-3} = -\frac{1}{3} \end{aligned}$$

14. If  $\alpha, \beta$  be roots of  $x^2 + px + 1 = 0$  and  $\gamma, \delta$  are the roots of  $x^2 + qx + 1 = 0$  then  $(\alpha - \gamma)(\beta - \gamma)(\alpha + \delta)(\beta + \delta)$  =

- (A)  $p^2 + q^2$  (B)  $p^2 - q^2$  (C)  $q^2 - p^2$  (D) None of these

यदि  $\alpha, \beta$  समीकरण  $x^2 + px + 1 = 0$  तथा  $\gamma, \delta$  समीकरण  $x^2 + qx + 1 = 0$  के मूल हों, तो  $(\alpha - \gamma)(\beta - \gamma)(\alpha + \delta)(\beta + \delta)$  =

- (A)  $p^2 + q^2$  (B)  $p^2 - q^2$  (C)  $q^2 - p^2$  (D) इनमें से कोई नहीं

**Ans.** C

**Sol.**  $x^2 + px + 1 = 0 \begin{cases} \alpha \\ \beta \end{cases} ; \quad x^2 + qx + 1 = 0 \begin{cases} \gamma \\ \delta \end{cases}$

$$\alpha + \beta = -p, \quad \alpha\beta = 1 ; \quad \gamma + \delta = -q, \quad \gamma\delta = 1$$

$$\begin{aligned}
& (\alpha - \gamma)(\beta - \gamma)(\alpha + \delta)(\beta + \delta) \\
&= (\alpha\beta - \alpha\gamma - \beta\gamma + \gamma^2)(\alpha\beta + \alpha\delta + \beta\delta + \delta^2) \\
&= (1 - \gamma(\alpha + \beta) + \gamma^2)(1 + \delta(\alpha + \beta) + \delta^2) \\
&= (1 + p\gamma + \gamma^2)(1 - p\delta + \delta^2) \\
&= 1 - p\delta + \delta^2 + p\gamma - p^2\gamma\delta + p\gamma\delta^2 + \gamma^2 - p\gamma^2\delta + \gamma^2\delta^2 \\
&= 1 - p\delta + \delta^2 + p\gamma - p^2 + p\delta + \gamma^2 - p\gamma + 1 \\
&= 2 - p^2 + \gamma^2 + \delta^2 \\
&= 2 - p^2 + (\gamma + \delta)^2 - 2\gamma\delta \\
&= 2 - p^2 + (-q)^2 - 2 \times 1 \\
&= 2 - p^2 + q^2 - 2 = q^2 - p^2
\end{aligned}$$

15. If a, b, c are real numbers satisfying the condition  $a + b + c = 0$  then the roots of the quadratic equation  $3ax^2 + 5bx + 7c = 0$  are :

(A) positive (B) negative (C) real and distinct (D) imaginary

यदि a, b, c वास्तविक संख्यायें शर्त  $a + b + c = 0$  को संतुष्ट करती हैं, तो द्विघात समीकरण  $3ax^2 + 5bx + 7c = 0$  के मूल :

(A) धनात्मक (B) ऋणात्मक (C) वास्तविक एवं भिन्न (D) काल्पनिक

**Ans.** C

**Sol.** If  $a + b + c = 0$

$$b = -(a + c)$$

$$b^2 = (a^2 + c^2 + 2ac)$$

$$\text{And Quadratic Equation : } 3ax^2 + 5bx + 7c = 0$$

$$D = 25b^2 - 84ac$$

$$D = 25(a^2 + c^2 + 2ac) - 84ac$$

$$D = 25a^2 + 25c^2 + 50ac - 84ac$$

$$D = 25a^2 + 25c^2 - 34ac$$

$$\therefore \frac{25a^2 + 25c^2}{2} \geq (25a^2 \times 25c^2)^{\frac{1}{2}}$$

$$25a^2 + 25c^2 \geq 50ac$$

$$25a^2 + 25c^2 - 50ac \geq 0$$

$$\therefore 25a^2 + 25c^2 - 34ac \geq 0$$

$$D \geq 0$$

$\therefore$  Roots are real and distinct.

16. Let a, b, c be three real numbers such that  $a + b + c = 0$  and  $a^2 + b^2 + c^2 = 2$ . Then the value of  $(a^4 + b^4 + c^4)$  is equal to :

तीन वास्तविक संख्याये  $a, b, c$  इस प्रकार है कि  $a + b + c = 0$  तथा  $a^2 + b^2 + c^2 = 2$  है। तो  $(a^4 + b^4 + c^4)$  का मान बराबर होगा:

- (A) 2 (B) 5 (C) 6 (D) 8

**Ans.** A

**Sol.** Given  $a + b + c = 0$

and  $a^2 + b^2 + c^2 = 2$

$$(a^2 + b^2 + c^2)^2 = 4$$

$$a^4 + b^4 + c^4 + 2[a^2b^2 + b^2c^2 + c^2a^2] = 4 \quad \dots(i)$$

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2 \Sigma ab$$

$$0 = 2 + 2\Sigma ab$$

$$\Sigma ab = -1$$

$$(ab + bc + ca)^2 = a^2b^2 + b^2c^2 + c^2a^2 + 2(ab^2c + abc^2 + a^2bc)$$

$$(-1)^2 = a^2b^2 + b^2c^2 + c^2a^2 + 2abc(a + b + c)$$

$$\Rightarrow a^2b^2 + b^2c^2 + c^2a^2 = 1$$

$\therefore$  from (i)

$$a^4 + b^4 + c^4 + 2 \times 1 = 4$$

$$a^4 + b^4 + c^4 = 2$$

17. The number of solution of the equation  $e^{2x} + e^x + e^{-2x} + e^{-x} = 3(e^{-2x} + e^x)$  is :

- (A) 0 (B) 2 (C) 1 (D) more than 2

समीकरण  $e^{2x} + e^x + e^{-2x} + e^{-x} = 3(e^{-2x} + e^x)$  के हलों की संख्या क्या है :

- (A) 0 (B) 2 (C) 1 (D) 2 से अधिक

**Ans.** C

**Sol.**  $e^{2x} + e^x + e^{-2x} + e^{-x} = 3(e^{-2x} + e^x)$

put  $t = e^x$

$$t^2 + t + \frac{1}{t^2} + \frac{1}{t} = 3\left(\frac{1}{t^2} + t\right)$$

$$t^4 + t^3 + 1 + t = 3(1 + t^3); \quad t \neq 0$$

$$t^4 + t^3 - 3t^3 + t + 1 - 3 = 0$$

$$t^4 - 2t^3 + t - 2 = 0$$

$$t^3(t - 2) + 1(t - 2) = 0$$

$$(t - 2)(t^3 + 1) = 0$$

$$(t - 2)(t + 1)(t^2 - t + 1) = 0$$

$$t = 2, -1 \quad \downarrow$$

$$e^x = 2 \quad (D < 0)$$

$$x = \ln 2$$

Only one solution

18. Complete set of values of 'a' such that  $y = \frac{x^2 - x}{1 - ax}$  ( $x \in \mathbb{R}$ ) attains all real values is :

यदि  $y = \frac{x^2 - x}{1 - ax}$  ( $x \in \mathbb{R}$ ) का मान सभी वास्तविक संख्याओं को प्राप्त होता  $a$  का मान होगा :

- (A)  $[1, 4]$  (B)  $(0, 4]$  (C)  $(0, 1]$  (D)  $(1, \infty)$

**Ans.** D

**Sol.**  $y = \frac{x^2 - x}{1 - ax}$

$$y = \frac{x(x-1)}{(1-ax)}, a \neq 1$$

$$y - axy = x^2 - x$$

$$x^2 + x - axy - y = 0$$

$$x^2 + (1 - ay)x - y = 0$$

$$D \geq 0$$

$$(1 - ay)^2 + 4y \geq 0$$

$$1 + a^2y^2 - 2ay + 4y \geq 0$$

$$a^2y^2 + (4 - 2a)y + 1 \geq 0$$

$$D \leq 0$$

$$(4 - 2a)^2 - 4a^2 \leq 0$$

$$16 + 4a^2 - 16a - 4a^2 \leq 0$$

$$16(a - 1) \geq 0$$

$$a \geq 1$$

$$a \in (1, \infty) \quad \therefore a \neq 1$$

19. If the roots of the equation  $bx^2 + cx + a = 0$  be imaginary, then for all real values of  $x$ , the expression

$$3b^2x^2 + 6bcx + 2c^2 \text{ is :}$$

[AIEEE-2009]

- (A) Greater than  $4ab$  (B) Less than  $4ab$  (C) Greater than  $-4ab$  (D) Less than  $-4ab$

यदि समीकरण  $bx^2 + cx + a = 0$  के मूल काल्पनिक हो तो  $x$  के सभी वास्तविक मानों के लिए व्यंजक  $3b^2x^2 + 6bcx + 2c^2$  :

[AIEEE-2009]

- (A)  $4ab$  से अधिक है (B)  $4ab$  से कम है (C)  $-4ab$  से अधिक है (D)  $-4ab$  से कम है

**Ans.** C

**Sol.**  $bx^2 + cx + a = 0$

$$\text{Roots are imaginary } c^2 - 4ab < 0$$

$$f(x) = 3b^2x^2 + 6bcx + 2c^2$$

$$D = 36b^2c^2 - 24b^2c^2 = 12b^2c^2$$

$$\therefore 3b^2 > 0$$

$$\therefore f(x) \geq \left( -\frac{D}{4a} \right)$$

$$f(x) = -c^2$$

$$\text{Now } c^2 - 4ab < 0$$

$$c^2 < 4ab$$

$$-c^2 > -4ab$$

$$\therefore f(x) > -4ab.$$

20. Let for  $a \neq a_1 \neq 0$ ,  $f(x) = ax^2 + bx + c$ ,  $g(x) = a_1x^2 + b_1x + c_1$  and  $p(x) = f(x) - g(x)$ . If  $p(x) = 0$  only for  $x = -1$  and  $p(-2) = 2$ , then the value of  $p(2)$  is : [AIEEE-2011]

माना  $a \neq a_1 \neq 0$  के लिए,  $f(x) = ax^2 + bx + c$ ,  $g(x) = a_1x^2 + b_1x + c_1$  तथा  $p(x) = f(x) - g(x)$ . यदि केवल  $x = -1$  के लिए  $p(x) = 0$  हो, तथा  $p(-2) = 2$  है, तो  $p(2)$  का मान है : [AIEEE-2011]

- (A) 18 (B) 3 (C) 9 (D) 6

Ans. A

Sol.  $P(x) = 0 \Rightarrow f(x) = g(x) \Rightarrow ax^2 + bx + c = a_1x^2 + b_1x + C,$   
 $\Rightarrow (a - a_1)x^2 + (b - b_1)x + (c - c_1) = 0.$

It has only one solution  $x = -1$

$$\Rightarrow b - b_1 = a - a_1 + c - c_1 \quad \text{_____ (1)}$$

$$\text{vertex } (-1, 0) \Rightarrow \frac{b - b_1}{2(a - a_1)} = -1 \Rightarrow b - b_1 = 2(a - a_1) \quad \text{_____ (2)}$$

$$\Rightarrow f(-2) - g(-2) = 2 \Rightarrow 4a - 2b + c - 4a_1 + 2b_1 - c_1 = 2$$

$$\Rightarrow 4(a - a_1) - 2(b - b_1) + (c - c_1) = 2 \quad \dots (3)$$

$$\text{by (1), (2) and (3) } (a - a_1) = (c - c_1) = \frac{1}{2}(b - b_1) = 2$$

$$\text{Now } P(2) = f(2) - g(2) = 4(a - a_1) + 2(b - b_1) + (c - c_1) = 8 + 8 + 2 = 18$$

21. The equation  $e^{\sin x} - e^{-\sin x} - 4 = 0$  has : [AIEEE-2012]

- (A) exactly four real roots (B) Infinite number of real roots  
 (C) No real roots (D) Exactly one real root

समीकरण  $e^{\sin x} - e^{-\sin x} - 4 = 0$  के :

[AIEEE-2012]

- (A) मात्र चार वास्तविक मूल (B) अनन्त वास्तविक मूल हैं  
 (C) कोई वास्तविक मूल नहीं है (D) मात्र एक वास्तविक मूल हैं

Ans. C

Sol. Let  $e^{\sin x} = t \Rightarrow t^2 - 4t - 1 = 0 \Rightarrow t = \frac{4 \pm \sqrt{16 + 4}}{2} \Rightarrow t = e^{\sin x} = 2 \pm \sqrt{5}$

$$\Rightarrow e^{\sin x} = 2 - \sqrt{5}, e^{\sin x} = 2 + \sqrt{5} \Rightarrow e^{\sin x} = 2 - \sqrt{5} < 0, (\text{rejected})$$

$$\Rightarrow \sin x = \ln(2 + \sqrt{5}) > 1 (\text{rejected})$$

hence no solution

22. If  $\alpha$  and  $\beta$  are roots of the equation  $px^2 + qx + r = 0$ ,  $p \neq 0$ . If  $p, q, r$  are in AP and  $\frac{1}{\alpha} + \frac{1}{\beta} = 4$  then value of  $|\alpha - \beta|$  is : [JEE(Main)-2014]

यदि समीकरण  $px^2 + qx + r = 0$ ,  $p \neq 0$  के मूल  $\alpha$  तथा  $\beta$  हैं तथा  $p, q, r$  स.श्रे. में हैं,  $\frac{1}{\alpha} + \frac{1}{\beta} = 4$  तो  $|\alpha - \beta|$  का मान होगा : [JEE(Main)-2014]

- (A)  $\frac{\sqrt{34}}{9}$  (B)  $\frac{2\sqrt{13}}{9}$  (C)  $\frac{\sqrt{61}}{9}$  (D)  $\frac{2\sqrt{17}}{9}$

Ans. B

Sol.  $px^2 + qx + r = 0 \begin{matrix} \alpha \\ \beta \end{matrix}$  ;  $p, q, r \rightarrow \text{A.P.}$  ;  $2q = p + r$

$$\frac{1}{\alpha} + \frac{1}{\beta} = 4 \Rightarrow \frac{\alpha + \beta}{\alpha\beta} = 4 \Rightarrow \frac{-q}{r} = 4$$

$$q = -4r \quad \text{_____ (i)}$$

$$\therefore -8r = p + r$$

$$p = -9r \quad \text{_____ (ii)}$$

$$|\alpha - \beta| = \sqrt{(\alpha + \beta)^2 - 4\alpha\beta} = \sqrt{\frac{q^2}{p^2} - \frac{4r}{p}} \quad \text{by (i) और (ii) से}$$

$$= \frac{\sqrt{q^2 - 4pr}}{|p|} = \frac{\sqrt{16r^2 + 36r^2}}{|-9r|} = \frac{2\sqrt{13}}{9}$$

### ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

23. Which of the following statement(s) is/are True?

- (A) The equation  $\sqrt{x+1} + \sqrt{x-1} = 1$  has no real solution  
 (B) If  $0 < p < \pi$  then the quadratic equation,  $(\cos p - 1)x^2 + \cos px + \sin p = 0$  has real roots  
 (C) If  $2a + b + c = 0$  ( $c \neq 0$ ) then the quadratic equation,  $ax^2 + bx + c = 0$  has no root in  $(0, 2)$

- (D) If  $x$  and  $y$  are positive real numbers and  $m, n$  are any positive integers then;  $\frac{x^n \cdot y^m}{(1 + x^{2n})(1 + y^{2m})} > \frac{1}{4}$

निम्नलिखित कथनों में से कौनसा कथन सत्य है?

- (A) समीकरण  $\sqrt{x+1} + \sqrt{x-1} = 1$  का कोई वास्तविक हल नहीं है  
 (B) यदि  $0 < p < \pi$  है तो द्विघात समीकरण  $(\cos p - 1)x^2 + \cos px + \sin p = 0$  के वास्तविक मूल हैं  
 (C) यदि  $2a + b + c = 0$  ( $c \neq 0$ ) तो द्विघात समीकरण  $ax^2 + bx + c = 0$  के अन्तराल  $(0, 2)$  में कोई वास्तविक मूल नहीं है

(D) यदि  $x$  तथा  $y$  दो धनात्मक संख्याये है तथा  $m, n$  दो धनात्मक पूर्णांक है तो;  $\frac{x^n \cdot y^m}{(1+x^{2n})(1+y^{2m})} > \frac{1}{4}$

**Ans.** AB

**Sol.** (A) Equation  $\sqrt{x+1} + \sqrt{x-1} = 1$

$$x \geq -1, x \geq 1$$

$$\text{Now } \sqrt{x+1} + \sqrt{x-1} = 1$$

$$x+1+x-1+2\sqrt{x^2-1} = 1$$

$$2\sqrt{x^2-1} = (1-2x)$$

$$4(x^2-1) = 1+4x^2-4x$$

$$4x^2-4 = 1+4x^2-4x$$

$$4x = 5$$

$$x = \frac{5}{4}$$

$$\text{at } \sqrt{1+\frac{5}{4}} + \sqrt{\frac{5}{4}-1}$$

$$= \frac{3}{2} + \frac{1}{2} = \frac{4}{2} = 2 \neq 1$$

$\therefore$  There is no real solution.

(B) if  $0 < p < \pi$

$$(\cos p - 1)x^2 + \cos px + \sin p = 0$$

$$D = \cos^2 p - 4 \sin p (\cos p - 1)$$

$$D = \cos^2 p - 4 \sin p \cos p + 4 \sin p$$

$$D = \cos^2 p + 4 \sin p (1 - \cos p)$$

$$D = \cos^2 p + 4 \sin p \cdot 2 \sin^2 \frac{p}{2}$$

$$D = \cos^2 p + 8 \sin p \cdot \sin^2 \frac{p}{2}$$

+ve +ve

$$\therefore D \geq 0$$

(C) if  $2a + b + c = 0$ , ( $c \neq 0$ )

Quadratic Equation  $ax^2 + bx + c = 0$

$$\text{Let } f(x) = ax^2 + bx + c$$

$$f(0) = 0 + 0 + c = c$$

$$f(2) = 4a + 2b + c$$

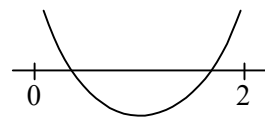
$$= 2(2a + b) + c$$

$$= 2(-c) + c$$

$$= -2c + c$$

$$= -c$$

here  $f(0)$ ,  $f(2)$  will be in same sign



$$(D) \frac{x^n \cdot y^m}{(1+x^{2n})(1+y^{2m})} > \frac{1}{4}$$

$$4x^n y^m > 1 + x^{2n} + y^{2m} + x^{2n} \cdot y^{2m}$$

$$x^{2n} + y^{2m} + 1 + x^{2n} \cdot y^{2m} - 4x^n y^m < 0$$

$$x^{2n} + y^{2m} - 2x^n \cdot y^m + 1 + x^{2n} \cdot y^{2m} - 2x^n y^m < 0$$

$$(x^n - y^m)^2 + (1 - x^n \cdot y^m)^2 < 0$$

$$\begin{matrix} +ve & & +ve & & \uparrow \end{matrix}$$

24. Let  $(a-1)(x^2 + \sqrt{3}x + 1)^2 - (a+1)(x^4 - x^2 + 1) \leq 0 \forall x \in \mathbb{R}$ , then which of the following is/are correct?

$$(A) a \in \left[-\frac{1}{\sqrt{3}}, \frac{4}{\sqrt{3}}\right]$$

(B) Largest possible value of  $a$  is  $\sqrt{3}$

(C) Number of possible integral values of  $a$  is 3 (D) Sum of all possible integral values of  $a$  is '0'

माना  $(a-1)(x^2 + \sqrt{3}x + 1)^2 - (a+1)(x^4 - x^2 + 1) \leq 0 \forall x \in \mathbb{R}$ , तो निम्न में से कौन सत्य है?

$$(A) a \in \left[-\frac{1}{\sqrt{3}}, \frac{4}{\sqrt{3}}\right]$$

(B)  $a$  का अधिकतम सम्भावित मान  $\sqrt{3}$  है

(C)  $a$  के सम्भावित पूर्णांक मानों की संख्या 3 है

(D)  $a$  के सभी सम्भावित पूर्णांक मानों का योग शून्य है

Ans. CD

$$\text{Sol. } (a-1)(x^2 + \sqrt{3}x + 1)^2 - (a+1)(x^4 - x^2 + 1) \leq 0$$

$$(a-1)(x^2 + \sqrt{3}x + 1)^2 - (a+1)(x^2 + \sqrt{3}x + 1)(x^2 - \sqrt{3}x + 1) \leq 0$$

$$\therefore x^4 - x^2 + 1 = (x^2 + \sqrt{3}x + 1)(x^2 - \sqrt{3}x + 1)$$

$$\Rightarrow (x^2 + \sqrt{3}x + 1) \left[ (a-1)(x^2 + \sqrt{3}x + 1) - (a+1)(x^2 - \sqrt{3}x + 1) \right] \leq 0$$

↓

$D < 0$ , always positive (removed)

$$\Rightarrow a(x^2 + \sqrt{3}x + 1) - a(x^2 - \sqrt{3}x + 1) - (x^2 + \sqrt{3}x + 1) - (x^2 - \sqrt{3}x + 1) \leq 0$$

$$a(x^2 + \sqrt{3}x - x^2 + \sqrt{3}x - 1) - (x^2 + \sqrt{3}x + 1 + x^2 - \sqrt{3}x + 1) \leq 0$$

$$2\sqrt{3}ax - 2x^2 - 2 \leq 0$$

$$x^2 - \sqrt{3}ax + 1 \geq 0$$

$\therefore a > 0$  then  $D \leq 0$  (should be)

$$\therefore 3a^2 - 4 \leq 0$$

$$(\sqrt{3}a + 2)(\sqrt{3}a - 2) \leq 0$$

$$a \in \left[ -\frac{2}{\sqrt{3}}, \frac{2}{\sqrt{3}} \right]$$

$$a = -1, 0, 1$$

$$\text{and sum of integral value} = -1 + 0 + 1 = 0$$

C, D options are right.

25. Let  $f(x) = x^2 + ax + b$ . If the maximum and the minimum values of  $f(x)$  are 3 and 2 respectively for  $0 \leq x \leq 2$ , then the possible ordered pair(s) of  $(a, b)$  is/are :

माना  $f(x) = x^2 + ax + b$  है। यदि  $f(x)$  का,  $0 \leq x \leq 2$  के लिये, अधिकतम एवं न्यूनतम मान क्रमशः 3 एवं 2 है, तो  $(a, b)$  के संभव क्रमित युग्म/युग्मों की संख्या है :

- (A)  $(-2, 3)$  (B)  $(-3/2, 2)$  (C)  $(-5/2, 3)$  (D)  $(-5/2, 2)$

**Ans.** A

**Sol.**  $f(x) = x^2 + ax + b$

$$\text{for } x \in [0, 2]$$

$$f(x) \in [2, 3]$$

$$f_{\max}(0) = b = 3$$

$$f_{\min} = 2$$

$$-\frac{D}{4a} = 2$$

$$\frac{4ac - b^2}{4a} = 2$$

$$\frac{4 \times 1 \times b - a^2}{4} = 2$$

$$12 - a^2 = 8$$

$$a^2 = 4$$

$$a = \pm 2$$

$$-\frac{b}{2a} = \frac{-a}{2} \Rightarrow \text{for } a = 2$$

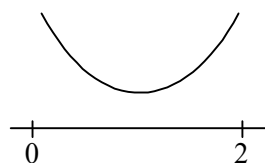
$$\frac{2}{2} = -1 \text{ (wrong)}$$

$$\text{for } a = -2$$

$$\frac{+2}{2} = 1 \text{ (right)}$$

$$a = -2$$

$$(a, b) = (-2, 3)$$



26. If the expression  $y = \frac{x-k}{x^2 - 4x + k}$  can take all real values for permissible real values of  $x$ , then the possible integral value of  $k$  can be :

यदि व्यंजक  $y = \frac{x-k}{x^2-4x+k}$   $x$  के अनुमेय (permissible) वास्तविक मानों के लिये सभी वास्तविक मान ले सकता है, तो  $k$  का संभव

पूर्णांक मान हो सकता है :

(A) 0

(B) 1

(C) 2

(D) 3

**Ans.** BC

**Sol.** If the expression  $y = \frac{x-k}{x^2-4x+k}$  can take all real values for permissible real values of  $x$

$$\Rightarrow y = \frac{x-k}{x^2-4x+k}$$

$$yx^2 - 4xy + ky - x + k = 0$$

$$yx^2 - x(4y+1) + ky + k = 0$$

$$y \neq 0, \quad D \geq 0$$

$$(4y+1)^2 - 4y(ky+k) \geq 0$$

$$16y^2 + 1 + 8y - 4ky^2 - 4ky \geq 0$$

$$(16-4k)y^2 + (8-4k)y + 1 \geq 0$$

for this

$$(i) a > 0$$

$$16-4k > 0$$

$$4k < 16$$

$$k < 4$$

$$(ii) D < 0$$

$$(8-4k)^2 - 4(16-4k) < 0$$

$$16(2-k)^2 - 16(4-k) < 0$$

$$4 + k^2 - 4k - 4 + k < 0$$

$$k^2 - 3k < 0$$

$$k(k-3) < 0$$

$$k \in (0, 3)$$

$$(i) \cap (ii)$$

$$k \in (0, 3)$$

$$k = 1, 2$$

B, C options are right.